Required Inputs for Hackathon

Use agentic AI and the GEMINI geothermal digital twin to select, size and validate a deep-geothermal system serving an annual heating and cooling load. The design must satisfy explicit thermal, hydraulic and operational constraints and explain every major engineering decision.

1 Objective and scope

Participants must build an agentic AI orchestration workflow that uses GEMINI for applicable subsurface/well-engineering evaluations and complementary component models for heating and cooling. The agent must interpret the provided demand profiles, generate system configurations, run engineering checks, detect violations, and iteratively improve a proposed design.

GEMINI repository (development branch):

https://github.com/GEMINI-Digital-Twin/gemini-geothermal-suite-open/tree/development

Scope clarification: the autonomous system-level design agent is to be built by participants; do not assume that GEMINI already contains an end-to-end heating-and-cooling optimizer.

2 Supplied annual demand profiles

Input dataset	File / format	Key figures
Heating demand	gemini_heating_demand_daily_2027.csv • date, heating_demand_MWth	2027 daily; max 30.0 MWth; 111.67 GWhth/year
Cooling demand	gemini_cooling_demand_daily_2027.csv • date, cooling_demand_MWcool	2027 daily; max 10.0 MWcool; 23.74 GWhcool/year

Each file contains 365 daily-average power values for 1 January to 31 December 2027. Heating peaks on 28 January; cooling peaks on 19 July. Profiles are synthetic and include seasonal variation and short demand excursions. Convert a day's power (MW) to daily energy (MWh) by multiplying by 24 hours.

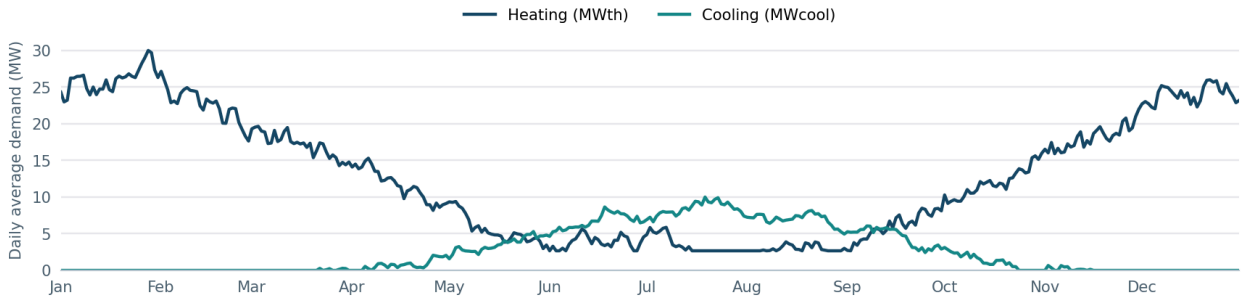


Figure 1. Daily synthetic heating and cooling demand throughout 2027. Source data are the two accompanying CSV files.

3 Fixed technical inputs

3.1 Geothermal reservoir

Parameter	Value	Unit / interpretation
Reservoir midpoint depth	2,500	m TVD
Reservoir gross thickness	100	m
Geothermal gradient	30	°C/km
Reference surface temperature	12	°C
Reservoir pressure	250	bar(g)
Permeability	200	mD

For selecting different geothermal well depth, you can use hydrostatic pressure for reservoir pressure and estimate reservoir temperature based on the provided geothermal gradient.

3.2 Geothermal brine and associated gas

Parameter	Specified value	Unit / note
Brine density	1,030	kg/m^3; reference constant
Dynamic viscosity	0.40	mPa·s; reference constant
Gas–liquid ratio	0.8	Nm^3 gas per m^3 produced liquid
Methane fraction in produced gas	80	mol% CH_4
Carbon dioxide fraction in produced gas	20	mol% CO_2
Fluid composition	Fixed	baseline; variable PVT optional

Define normal cubic metres (Nm^3) at 0 °C and 1.01325 bar(a). GLR refers to potentially liberated gas at standard conditions.

3 Design and operating constraints

Constraint	Required setting	Status / interpretation
Number of geothermal doublets	Choose 1–3	design variable
Maximum production flow	300 m^3/h per doublet	hard challenge cap
Minimum stable flow while active	50 m^3/h per doublet	hard challenge cap
Maximum geothermal brine cooling	40 °C	hard challenge cap; producer to injector
Pump hydraulic efficiency	70%	simplified
Production / injection volume balance	Required	same reservoir; equal liquid volumes
Annual operation of geothermal doublet	330 daily steps	daily power and energy balance
Supply temperature to the demand (heating)	100 °C	fixed
Supply temperature to the demand (Cooling)	20 °C	fixed